

Coarsening and Arrest in Colloidal Gels: Modelling via Brownian Dynamics

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Contents

1	Introduction	2
2	Results	2
2.1	Particle configurations	2
2.2	Structure factor	2
2.3	Ageing and MSD	2

1 Introduction

This report investigates coarsening and arrest in colloidal gels using Brownian Dynamics simulations.

2 Results

2.1 Particle configurations

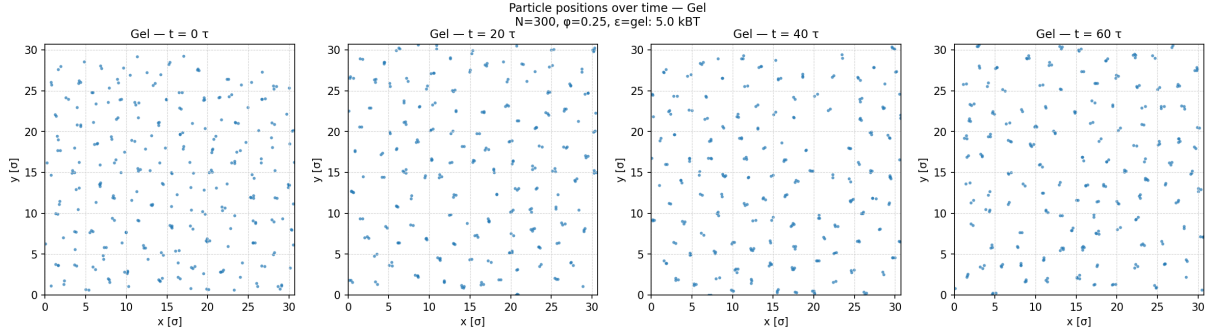


Figure 1: Particle configurations for the gel.

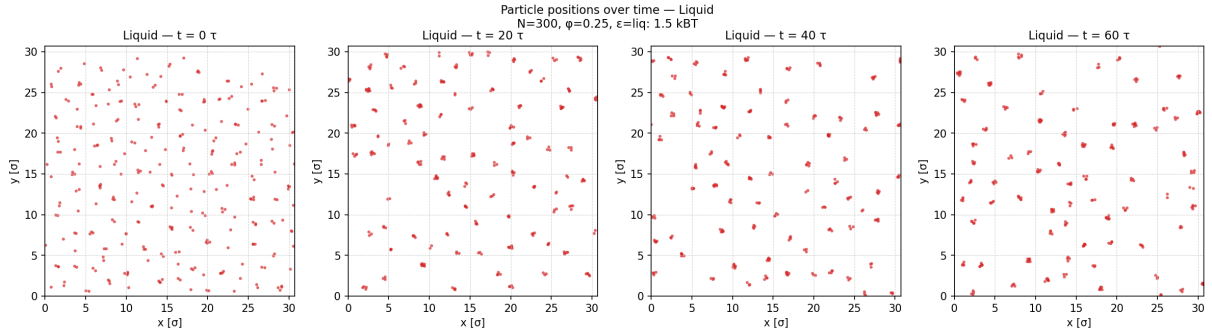


Figure 2: Particle configurations for the liquid reference.

2.2 Structure factor

2.3 Ageing and MSD

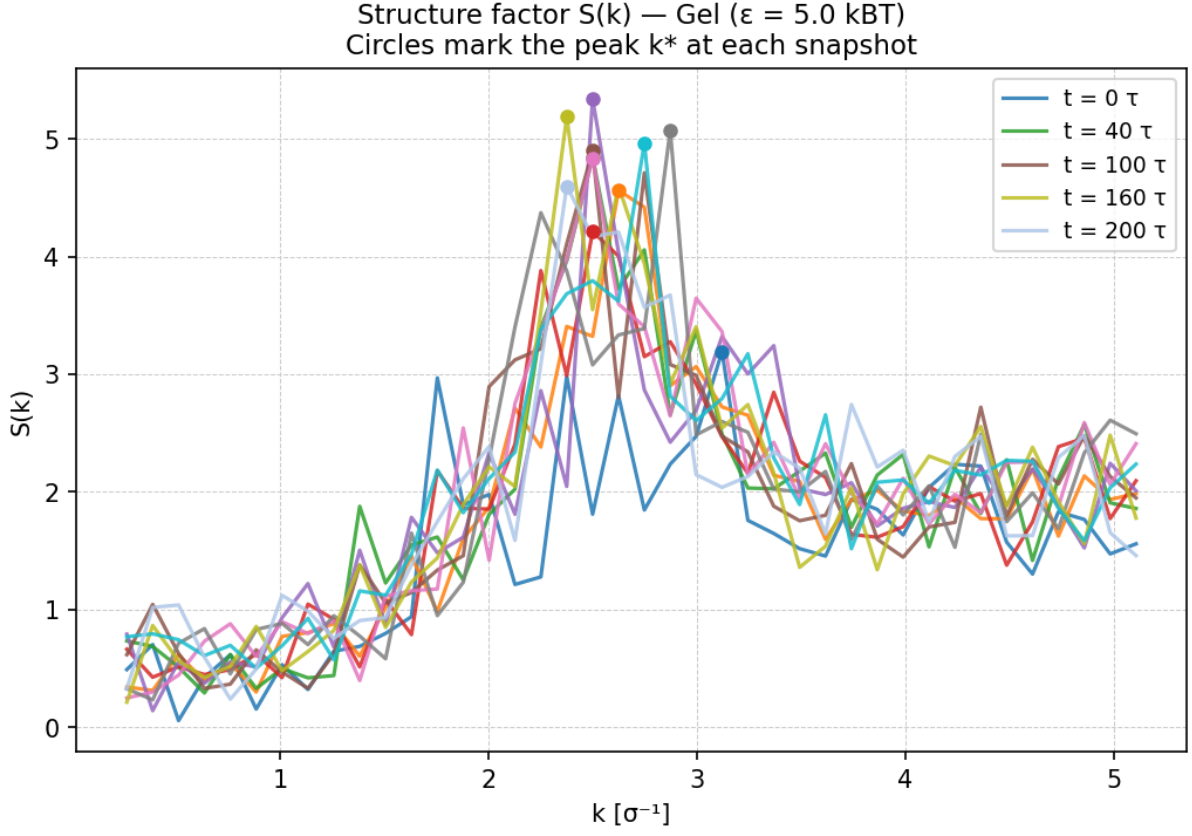


Figure 3: Structure factor evolution in the gel.

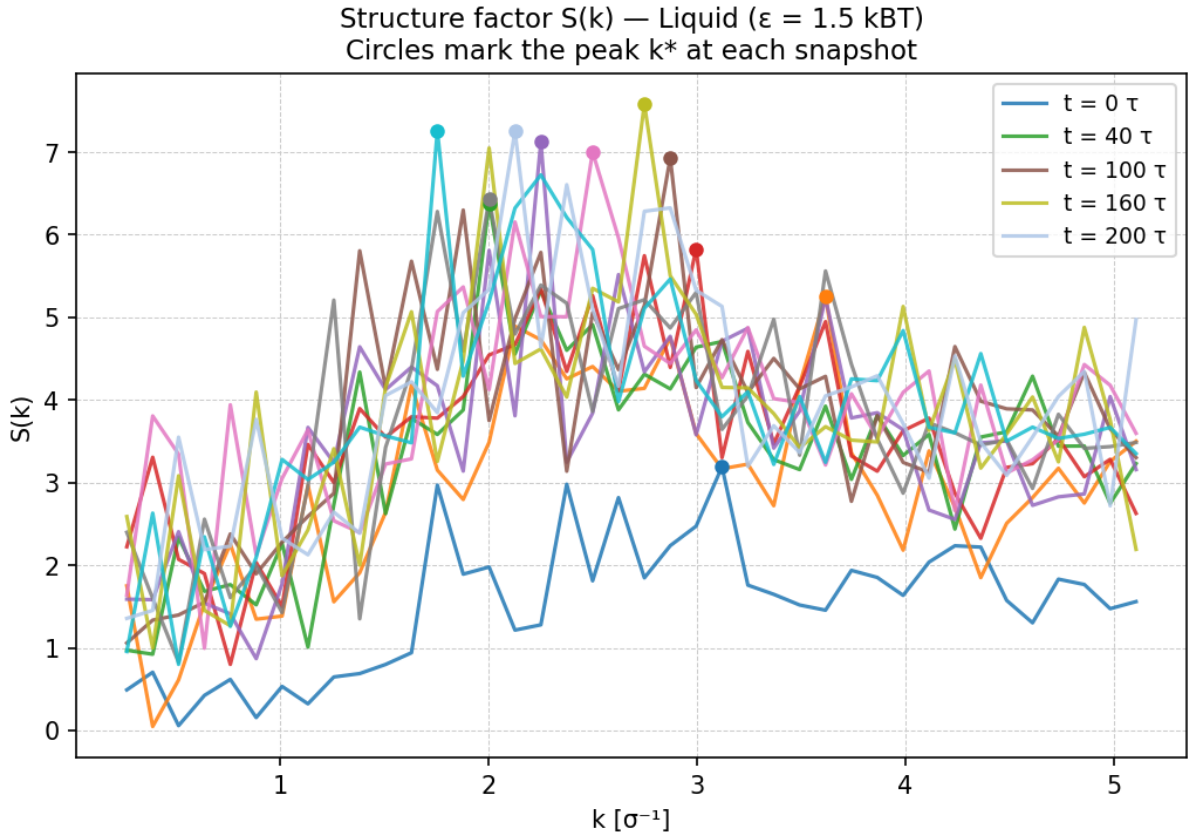


Figure 4: Structure factor evolution in the liquid.

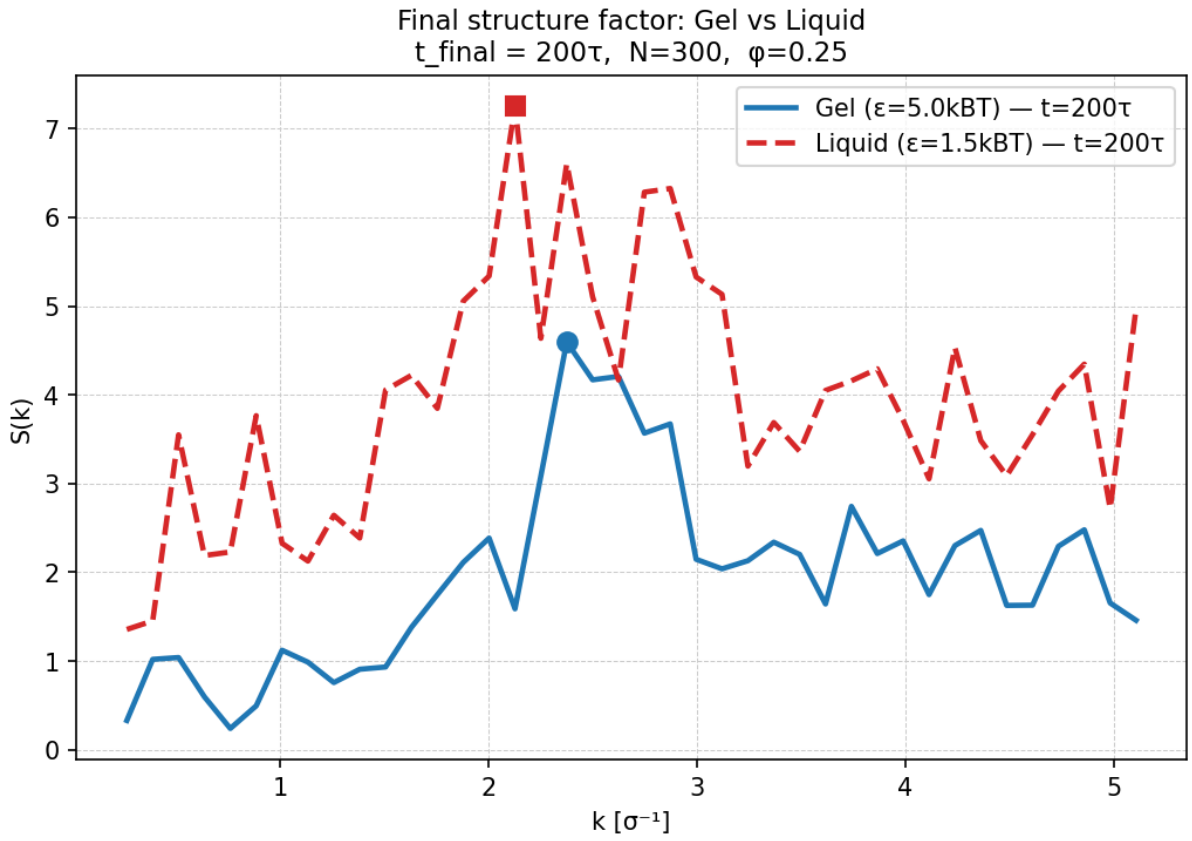


Figure 5: Final structure factor comparison.

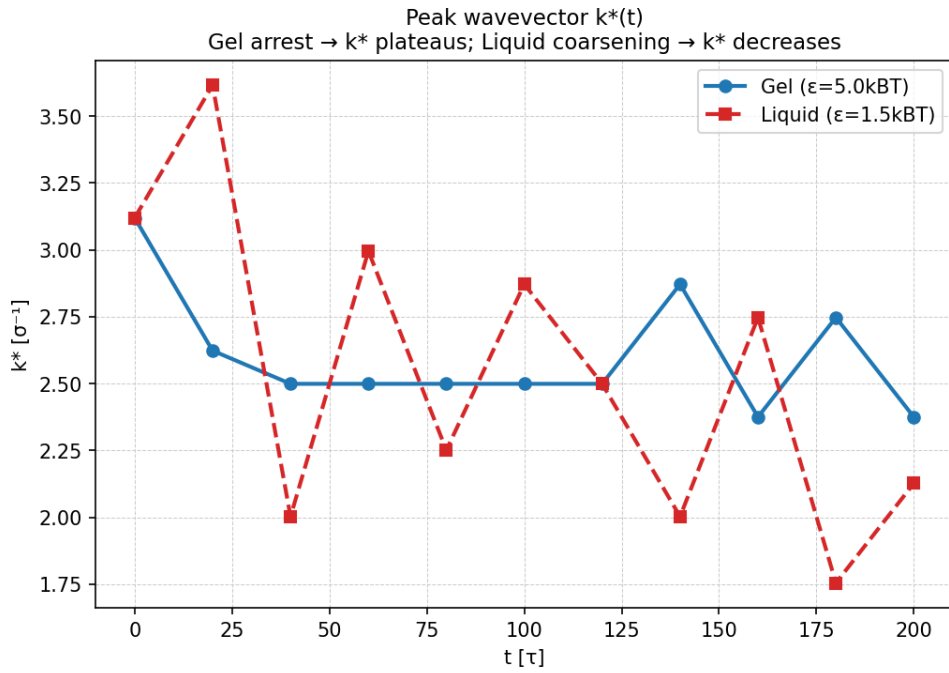


Figure 6: Peak wavevector $k^*(t)$.

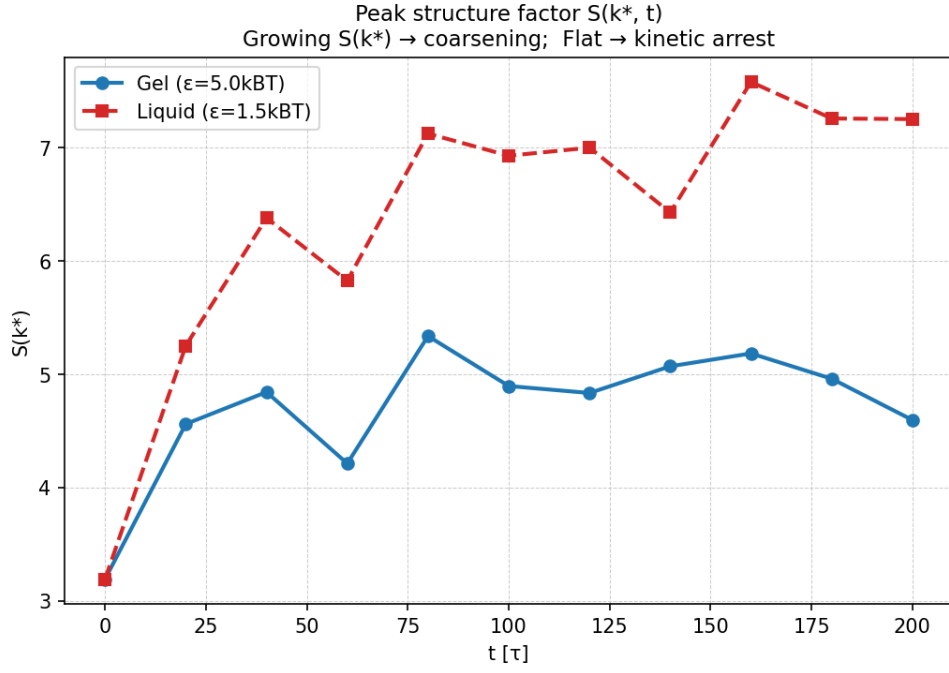


Figure 7: Peak structure factor amplitude.

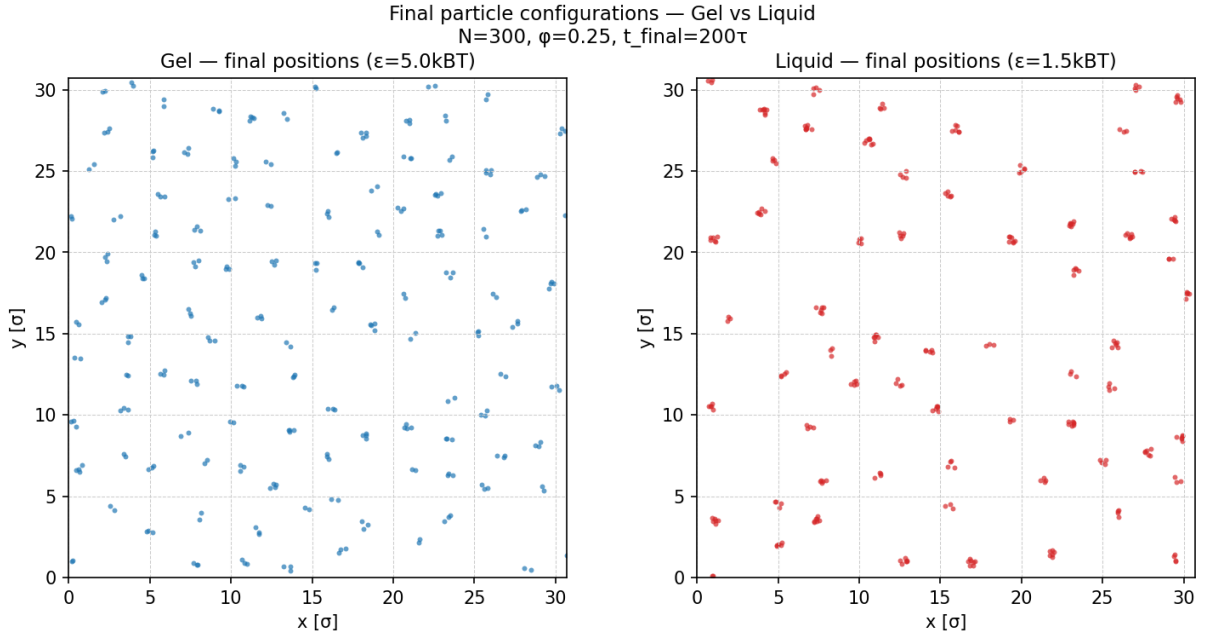


Figure 8: Final particle configurations.

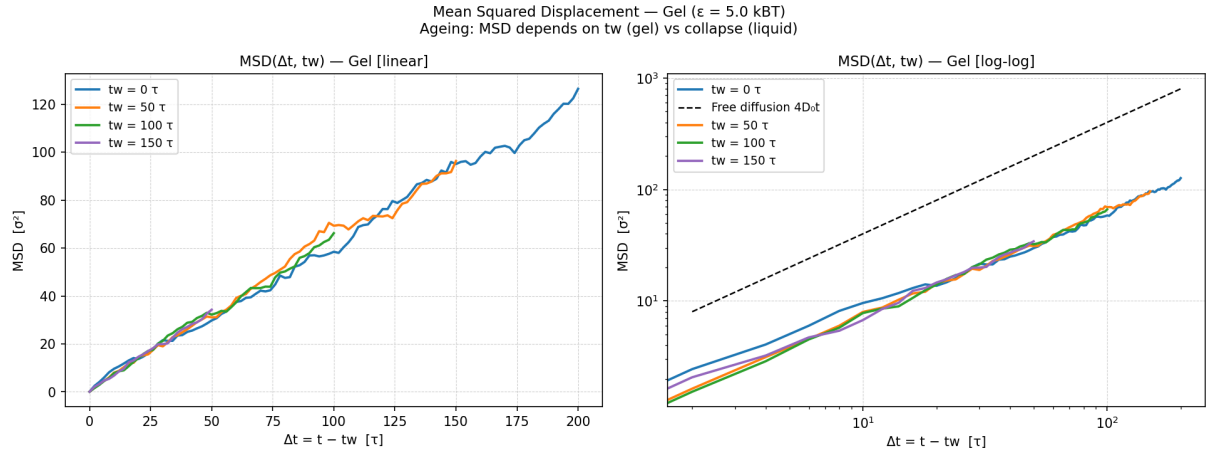


Figure 9: MSD in the gel.

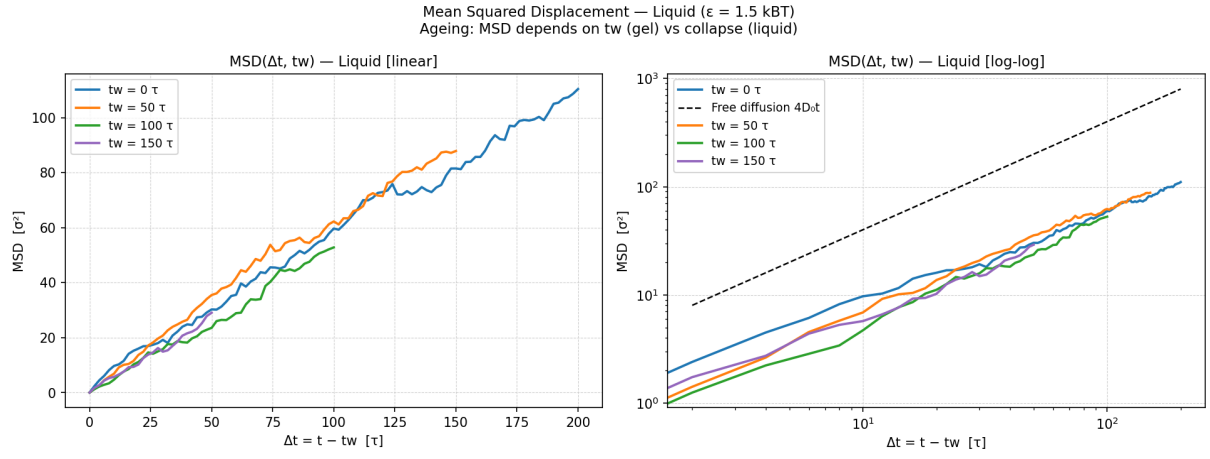


Figure 10: MSD in the liquid.

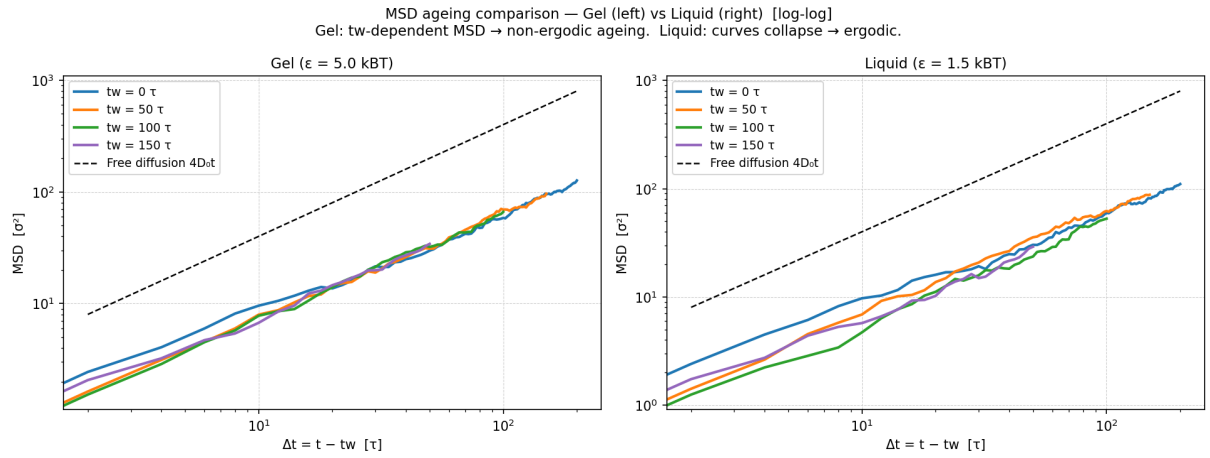


Figure 11: Ageing comparison.

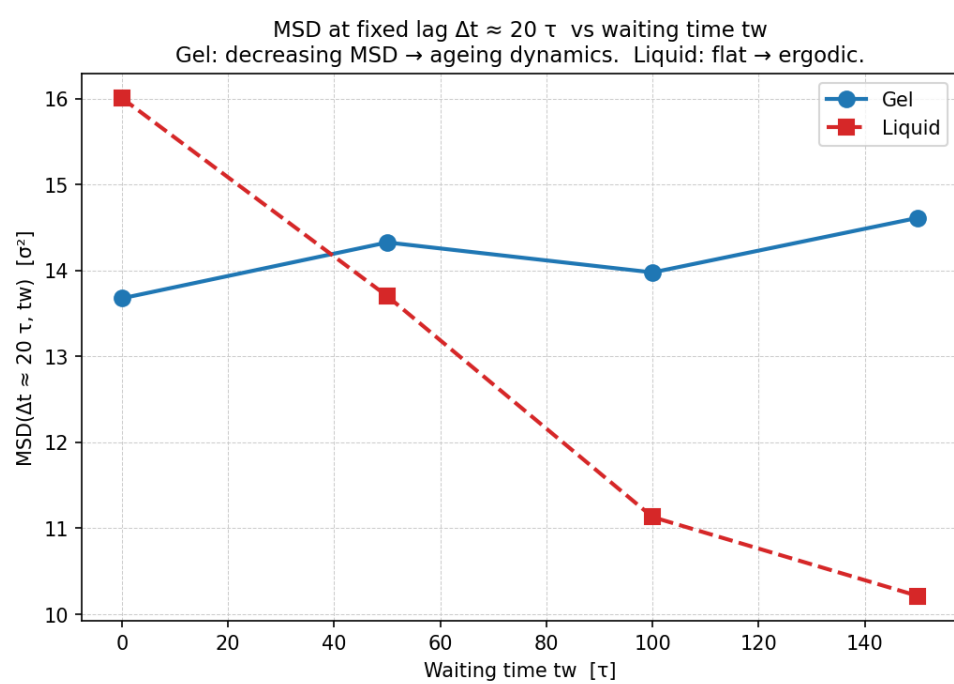


Figure 12: MSD versus waiting time.